



Date: Aug 28, 2025

Math Section 1.1

Practice Test 1

Name:

Class of:

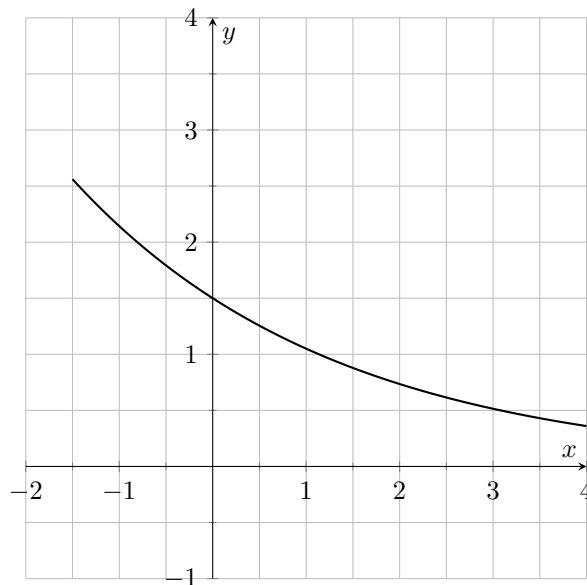
Teacher:

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Solutions

1. Graph Interpretation

Prompt. Based on the graph, which of the following statements are true? Select ALL that apply.



1. $a > 0$
2. $a < 0$
3. $b > 1$
4. $0 < b < 1$

Correct choices: A, D

Solution. For $f(x) = ab^x$, the y -intercept is $f(0) = a$. From the graph $f(0) > 0 \Rightarrow a > 0$ (A true, B false). The curve decreases and levels toward $y = 0$, so $0 < b < 1$ (D true, C false).

2. Table Interpretation

Prompt. Which of the following statements about the function described by the table are true? Select ALL that apply.

x	$f(x)$
1	6
5	96
8	768

1. This is an exponential decay function.
2. The initial value is positive.
3. There is an asymptote at $y = 0$.
4. There is not enough information to determine whether any of the above statements are true.

Correct choices: B, C

Solution. Find the base using equal-step ratios:

$$\frac{f(5)}{f(1)} = \frac{96}{6} = 16 = b^4 \Rightarrow b = 2, \quad \frac{f(8)}{f(5)} = \frac{768}{96} = 8 = b^3 \Rightarrow b = 2 (\checkmark).$$

Thus $b = 2 > 1$ (growth, so A is false). Solve for a with $x = 1$: $a \cdot 2 = 6 \Rightarrow a = 3 > 0$ (B true). Any $a b^x$ has horizontal asymptote $y = 0$ (C true). The data are sufficient (D false).

3. Solve an Exponential Equation

Prompt. What is the value of t in the following equation?

$$3^{5t-1} = 9^{t+4}$$

1. $\frac{11}{3}$
2. $\frac{7}{2}$
3. -5
4. 3

Correct choice: D

Solution. Rewrite $9 = 3^2$: $9^{t+4} = 3^{2(t+4)} = 3^{2t+8}$. Since bases match and are $> 0 \neq 1$, set exponents equal:

$$5t - 1 = 2t + 8 \Rightarrow 3t = 9 \Rightarrow \boxed{t = 3}.$$

4. Percent Depreciation from a Base

Prompt. The value $v(t) = P(0.92)^t$. By what percentage does the value depreciate each year?

1. 0.08%
2. 0.92%
3. 8%
4. 92%

Correct choice: C

Solution. Write $0.92 = 1 - r \Rightarrow r = 1 - 0.92 = 0.08 = 8\%$. So the car loses 8% per year.

5. Equivalent Exponential Forms

Prompt. $P(t) = 5(1.9)^{3t}$. Which equations are equivalent? Select ALL that apply.

$$\text{A } 5(3.8)^t \quad \text{B } 5[(1.9)^t]^3 \quad \text{C } 5(1.9)(1.9)^t \quad \text{D } 5(6.859)^t$$

Correct choices: B, D

Solution. Use $a^{3t} = (a^t)^3 = (a^3)^t$.

$$5(1.9)^{3t} = 5((1.9)^t)^3 = 5((1.9)^3)^t = 5(6.859)^t.$$

A multiplies the base by 2 (not valid). C equals $5(1.9)^{t+1}$ (changes the exponent).

6. Half-Life Reasoning (Free Response)

Prompt. A medication has half-life 4 h. Explain why some of the original dose always remains, yet taking another dose after 12 h is safe.

Solution. Half-life model:

$$A(t) = A_0 \left(\frac{1}{2}\right)^{t/4}.$$

Since $A(t) > 0$ for all t and $A(t) \rightarrow 0^+$ as $t \rightarrow \infty$, a tiny amount always remains (horizontal asymptote $y = 0$). After 12 h there are $12/4 = 3$ half-lives, so

$$A(12) = A_0 \left(\frac{1}{2}\right)^3 = \frac{A_0}{8} = 12.5\% \text{ remaining.}$$

Because the residue keeps halving, the amount becomes negligible; clinical schedules treat re-dosing at 12 h as safe.